Integration with SymPy and SciPy

Exact and numerical integrals in one dimension

Sharareh Sayyad

Washington State University

October 30, 2026

Two routes to a definite integral

$$\int_a^b f(x)\,dx \;=\; F(b) - F(a), \qquad F'(x) = f(x)$$

Exact: SymPy

Find an antiderivative F.

✓ a formula, no rounding

✗ not every f has one

Numerical: SciPy

Add up f at sample points.

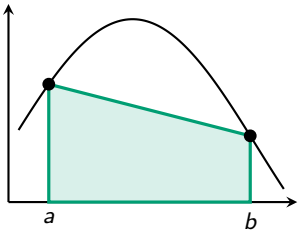
✓ always gives a number

✗ only approximate: check the error

Same idea as derivatives: **formula** vs. **data**.

Numerical integration: three rules

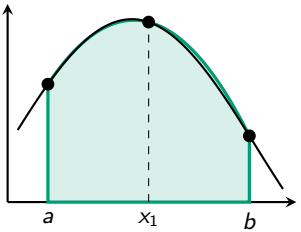
Trapezoidal rule



straight line through the ends

$$E \approx -\frac{(b-a)\,h^2}{12}\,f''(\xi)$$

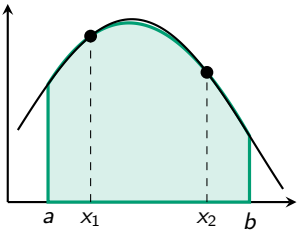
Simpson's rule



parabola through 3 points

$$E \approx -\frac{(b-a)\,h^4}{180}\,f^{(4)}(\xi)$$

Gauss quadrature



best points x_i and weights w_i

x_i: roots of Legendre P_n

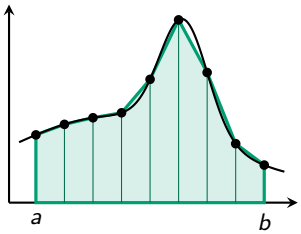
($n = 2$: $\pm 1/\sqrt{3}$ on $[-1, 1]$)

$$E = \frac{(b-a)^5}{4320}\,f^{(4)}(\xi) \quad (n = 2)$$

equal spacing $h = \frac{b-a}{n}$ vs. Gauss chooses where to sample

Many points: split $[a, b]$ into pieces

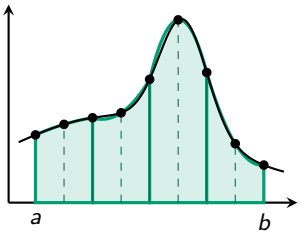
Trapezoid, $n = 8$



a line on each piece

$$\frac{h}{2}\left(f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n\right)$$

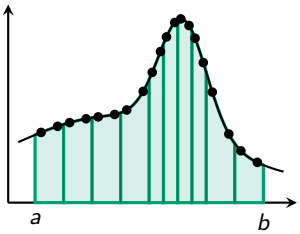
Simpson, $n = 8$



a parabola on each pair
so n must be **even**

$$\frac{h}{3}\left(f_0 + 4f_1 + 2f_2 + \cdots + f_n\right)$$

quad: adaptive Gauss



finer pieces near the peak

$$\sum_{i=1}^{n} w_i\, f(x_i)$$